Which ESM-2 layer should you feed a mutation-effect predictor? A 33-layer × 4-protein audit

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Abstract

Supervised mutation-effect predictors built on protein language models (PLMs) inherit a hidden degree of freedom: which transformer layer to read out. Probing studies in NLP and, more recently, in protein modelling suggest that intermediate layers carry the most structural and functional signal, while the final layers specialize toward the pre-training objective. Whether this picture governs deep-mutational-scanning (DMS) prediction is rarely tested directly; most pipelines simply default to the last layer. Here we audit the question exhaustively on four DMS systems (SARS-CoV-2 RBD binding, caspase-3 activity, a TIM-barrel fitness landscape, and the λ cI repressor) by training matched probe heads on all 33 layers of ESM-2 (650M), under two architectures — a single-stream baseline and a two-stream cross-attention model that fuses each layer with an alignment-free structural representation (Z_{II}, extracted from RoseTTAFold3) — with paired splits and paired initializations (33 layers × 3 splits × 2 seeds per cell). The optimal layer is protein-specific: RBD peaks at layer 10 (Spearman ρ 0.5141 versus 0.4536 at the last layer), cI at layer 32, and TIM and caspase-3 at the last layer. Fusing a structural second stream lifts accuracy but does not relocate the optimum (cross-layer correlation of the two profiles 0.78–0.98). The fusion gain is itself layer-structured, largest at early layers and shrinking to below the ± 0.02 reproducibility floor exactly where the PLM stream is already strongest. Linear CKA between each ESM-2 layer and the structural representation supports the "intermediate layers encode structure" picture on two proteins and inverts it on the other two, and an effective-rank analysis shows the last-layer representation collapses to roughly one quarter of the intrinsic dimensionality of the best intermediate layer. We conclude that layer choice is a dataset-specific hyperparameter that should be swept, not assumed, and we provide a cheap protocol for doing so.

1 Introduction

Protein language models (PLMs) trained on hundreds of millions of sequences produce per-residue representations that transfer broadly to structure and function prediction [1, 2]. In supervised variant-effect prediction, the dominant recipe is to freeze the PLM, read out one layer — almost always the last — and train a small prediction head on top [3]. The choice of layer is rarely discussed, yet it is a genuine degree of freedom: a 33-layer model such as ESM-2 650M offers 33 materially different representations of the same sequence. Two forces make the choice non-trivial. On one hand, deeper layers integrate more context and approach the representation used for the model's own pre-training predictions. On the other hand, the masked-modelling objective shapes the final layers toward token reconstruction, not toward any particular downstream quantity, so there is no guarantee that the last layer is the best read-out for a fitness label.

A loose consensus from probing work suggests where the useful signal should sit. In NLP, layer-wise probing showed that BERT's layers recapitulate a classical processing pipeline, with syntactic structure concentrated in intermediate layers [4]. Analyses of contextual embeddings further showed that final layers are highly anisotropic and specialized to the training objective [5]. For PLMs specifically, recent geometric analyses report that structure-related signal peaks before the final layer and that effective dimensionality contracts toward the output layers [6]. If this picture held universally for mutation-effect prediction, practitioners could fix an intermediate layer once and for all. Whether it holds is an empirical question, and it matters in practice: layer choice is made silently in every DMS pipeline, costs nothing to sweep once features are cached, and interacts with a second increasingly common design decision — whether to fuse PLM embeddings with structure-based representations.

We test this directly. Rather than probing for what a layer *encodes*, we measure what a layer *affords*: we train identical supervised heads on each of the 33 layers of ESM-2 650M across four deep-mutational-scanning (DMS) datasets spanning different proteins, assays, and selection types, with position-held-out validation and paired seeds. We then repeat the entire sweep in a two-stream architecture that fuses each layer with Z_{II}, an alignment-free structural representation extracted from RoseTTAFold3 (RF3) [8]. This design answers three practical questions: (i) does the best layer coincide across proteins; (ii) does adding structural information change which layer is best; and (iii) where along the depth axis does a structural second stream actually add information? We complement the supervised curves with layer-resolved representational-similarity (linear CKA [7]) and effective-rank measurements that connect the performance profile to the geometry of the representations.

Throughout, we interpret differences against an empirically calibrated noise floor of ± 0.02 in validation Spearman ρ (Methods), and we do not draw conclusions from differences smaller than this floor.

2 Results

2.1 The optimal ESM-2 layer is protein-specific

Figure 1 (grey curves) shows validation Spearman ρ of single-stream (SP) probe heads trained on each of the 33 ESM-2 layers for the four DMS systems, averaged over 3 position-held-out splits $\times$ 2 initializations. The four profiles share no common optimum (Table 1). RBD binding prediction peaks at layer 10 ($\rho = 0.5141$), exceeding the conventional last-layer choice by +0.0605 (0.4536 at L33) — three times the noise floor. The λ cI repressor peaks at L32 (0.5416 versus 0.5037 at L33, +0.0380). In contrast, TIM-barrel fitness and caspase-3 activity are predicted best from the last layer (0.5992 and 0.5826), and both profiles rise essentially monotonically with depth. The peaks are robust to resampling: the per-layer standard deviation across the six split $\times$ seed runs (median 0.026–0.061 across proteins) reflects split heterogeneity more than run-to-run jitter, and the RBD and CI peak-to-L33 gaps exceed it comfortably. Thus the "intermediate layers are best" expectation is confirmed in one system out of four, the "last layer" default is correct in two, and one system prefers the penultimate layer. Neither of the two standard heuristics — always use the last layer, or always use a fixed intermediate layer — is safe.

Beyond the peak positions, the profile *shapes* differ qualitatively. RBD rises from 0.366 at L1 to a broad early plateau ($\approx$ 0.49–0.51 over L5–L12) and then decays monotonically to the last layer, losing 0.06 from plateau to L33. CI, the smallest dataset ($N = 351$) and the noisiest profile (median per-layer s.d. 0.061), dips in the early-middle layers (0.285 at L5) and climbs steeply from L15 (0.459) to its L32 peak before dropping 0.038 at L33. TIM and CASP3 both show poor early layers ($\approx$ 0.30–0.36 through L8) followed by an essentially monotone climb (TIM: 0.488 at L10 to 0.599 at L33; CASP3: 0.421 at L10 to 0.583 at L33). The early-layer weakness of TIM and CASP3

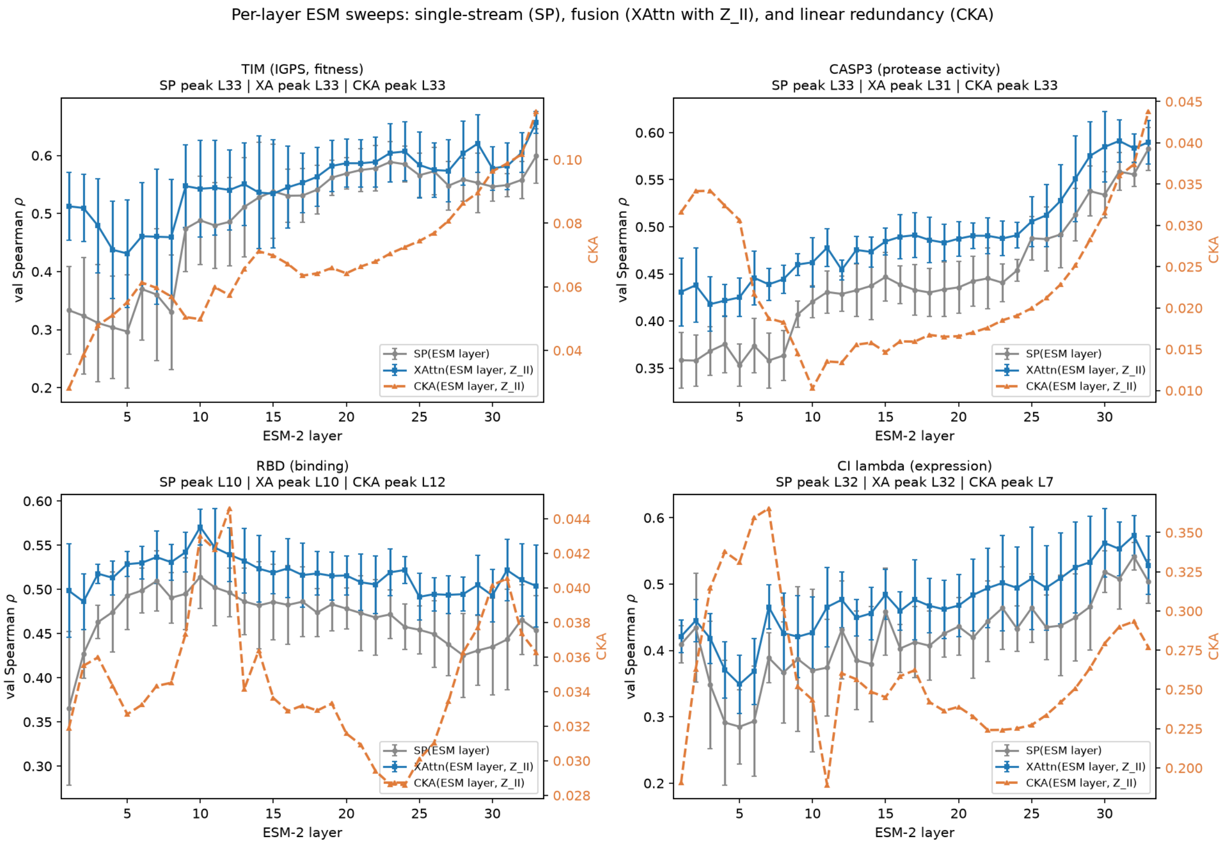

Figure 1: Per-layer ESM-2 sweeps on four DMS systems. Grey: single-stream (SP) probe validation Spearman ρ per layer (mean over 3 position-held-out splits $\times$ 2 initializations; error bars ± 1 s.d.). Blue: two-stream cross-attention fusion (XAttn) of the same layer with Z_{II}. Orange dashed (right axis): linear CKA between the layer's per-residue representation and Z_{II}. Panel headers report the peak layer of each curve.

foreshadows the fusion result below: it is exactly these poorly performing layers that gain most from a structural second stream.

2.2 Fusing a structural second stream does not relocate the optimal layer

Repeating the sweep with the XAttn fusion architecture (Fig. 1, blue curves) lifts performance on every protein and nearly every layer: the fusion marginal is positive on 131 of 132 protein $\times$ layer cells and exceeds the noise floor in 119 of them. The fusion profiles nevertheless track the SP profiles closely: the per-layer Pearson correlation between the SP and XAttn curves is 0.98 (CASP3), 0.95 (CI), 0.92 (TIM), and 0.78 (RBD). Accordingly, the XA optimum sits at the same layer as the SP optimum on RBD (L10), CI (L32), and TIM (L33), and on CASP3 it moves from L33 to L31 by +0.0015 — an order of magnitude below the noise floor and therefore not a real relocation. Layer choice and the fusion decision are thus approximately separable: structural context changes *how well* a layer performs, not *which* layer is best.

Table 1: Peak layers and last-layer baselines. Mean validation Spearman ρ (3 splits $\times$ 2 inits). SP = single-stream probe; XA = cross-attention fusion with Z_{II}; marginal = XA − SP at the SP-optimal layer. Δ_{L33} = SP best − SP L33. The ± 0.02 noise floor applies to all differences.

Protein	SP best (layer)	SP L33	Δ_{L33}	XA best (layer)	XA L33	marginal
RBD	0.5141 (L10)	0.4536	+0.0605	0.5706 (L10)	0.5038	+0.0565
CASP3	0.5826 (L33)	0.5826	0	0.5912 (L31)	0.5897	+0.0071
TIM	0.5992 (L33)	0.5992	0	0.6568 (L33)	0.6568	+0.0576
CI	0.5416 (L32)	0.5037	+0.0380	0.5729 (L32)	0.5283	+0.0313

2.3 The fusion gain is layer-structured

The per-layer fusion marginal (XA − SP on identical splits and seeds; Fig. 2) is far from constant (per-protein means +0.047 to +0.057), and its shape is itself system-specific. On TIM it is U-shaped: +0.179 at L1, decaying to −0.004 at L15 — statistically indistinguishable from zero — then recovering to +0.058 at L33. On RBD the marginal is largest at L1 (+0.133) and re-rises in deep layers (L31 +0.079) precisely where the SP profile decays, i.e. the structural stream rescues degenerating layers. On CASP3 the marginal declines with depth and reaches +0.007 at L33 — below the noise floor, meaning the structural stream adds nothing at CASP3's best layer. The pattern is consistent across systems: the second stream pays most where the PLM stream is intrinsically weak, and least where it is already optimal. Notably, the TIM marginal recovers at L33 exactly where ESM–Z_{II} linear similarity is maximal (Section 2.4), so high linear redundancy does not imply uselessness: the fused model still extracts non-linearly complementary signal there.

2.4 Representational similarity to structure: the literature picture holds in half the systems

To relate these performance profiles to the "intermediate layers encode structure" hypothesis, we measured linear CKA between each layer's per-residue representation and the Z_{II} structural representation (Fig. 1, orange curves). RBD conforms to the expected picture: CKA peaks at L12 (0.0446), adjacent to the SP performance peak at L10. CI also peaks early (L7, 0.365; we caution that the CI Z_{II} extraction carries a known indexing artifact, Methods). The other two systems invert the picture: TIM's CKA rises monotonically to its maximum at L33 (0.115), and CASP3's is U-shaped with a minimum at L9–15 (≈ 0.010) and its maximum, again, at L33 (0.0438). Consistent with this split, the rank correlation between per-layer CKA and per-layer SP performance is strongly positive on TIM ($\rho_s = +0.78$) and weak or negative elsewhere (CASP3 +0.22, RBD +0.14, CI −0.22). Structural alignment of intermediate layers is therefore a system-dependent property, not a universal feature of ESM-2 depth: the geometry that a probing study extracts from one protein family does not transfer as a rule to another.

2.5 Mechanistic clue: the last layer is intrinsically low-rank

Why would L10 outperform L33 on RBD? A per-representation effective-rank analysis (Fig. 3) shows that the RBD last-layer representation occupies far fewer directions than the best intermediate layer: participation ratio 44.5 at L33 versus 205.7 at L12 (4.6×), and 325 versus 702 singular values needed for 90% of variance. An independent probe comparison on RBD confirms that the difference is not an artifact of the main sweep: L12 scores 0.4955 against L33's 0.4821 under a matched protocol. This is consistent with reports that final layers of masked language models specialize toward the pre-training objective and become anisotropic [5, 6]: on systems where the

Figure 2: Fusion marginal per layer (XAttn − SP, paired splits and seeds). The shaded band marks the ±0.02 reproducibility floor: marginals inside it (e.g. CASP3 at L33, TIM near L15) are treated as zero.

assay rewards structure-shaped intermediate features, the collapsed final layer is simply the wrong read-out. We stress that this contraction is a property of the representation, not of the label: TIM and CASP3 achieve their best ρ on exactly this low-rank L33 read-out.

3 Discussion

Our audit yields a simple practical message: the ESM-2 layer fed to a mutation-effect predictor is a dataset-specific hyperparameter, and both popular defaults fail somewhere. "Always use the last layer" forfeits +0.06 Spearman on RBD and +0.04 on CI; "use a fixed intermediate layer" forfeits the monotonic gains on TIM and CASP3. The remedy is cheap. Because all 33 layers come out of a single forward pass, the sweep costs one feature extraction plus 33×6 short head-trainings on frozen embeddings — minutes per layer on a single consumer GPU at our dataset sizes, orders of magnitude less than any fine-tuning step. We therefore recommend a single-stream coarse sweep over all layers as a standard first step in DMS pipelines, with the winner confirmed under the intended final architecture. Two softer conclusions qualify the fusion literature. First, an auxiliary structural stream does not relocate the optimal PLM layer, so layer selection and multimodal design can be tuned independently. Second, the value of an auxiliary stream is predictable in sign and rough location from the single-stream profile itself: it pays where the PLM representation is poor (early layers, or deep layers whose single-stream performance has decayed) and vanishes —

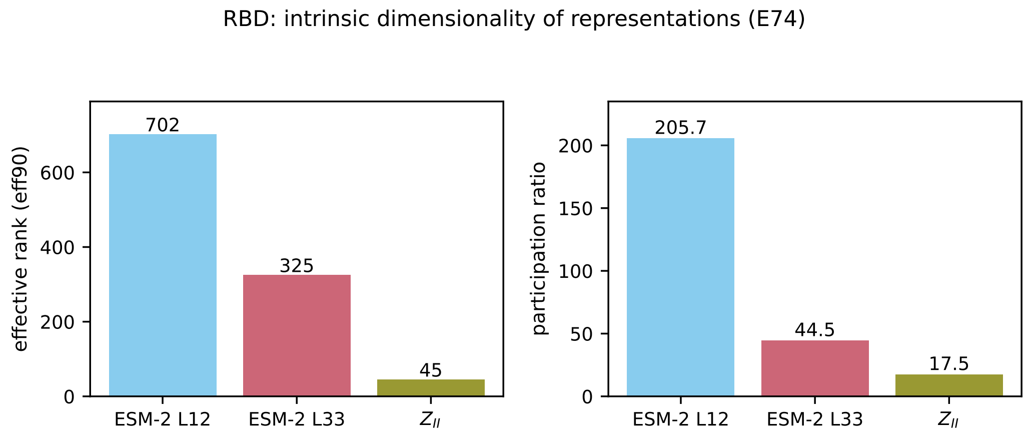

Figure 3: Intrinsic dimensionality of RBD representations. Effective rank (left: number of singular values capturing 90% of variance; right: participation ratio) of ESM-2 L12, ESM-2 L33, and Z_{II}, computed over per-residue vectors.

below a ± 0.02 reproducibility floor — where the PLM is already at its best. Practitioners fusing structure with PLM features should therefore not expect uniform additive gains across depths, and a single-layer fusion benchmark can misrepresent what fusion offers at other depths.

Our results also draw a line between two questions that are easy to conflate. Probing studies ask which layers *contain* structure- or function-related signal, typically with linear read-outs and information-theoretic controls; our audit asks which layer *serves* a supervised mutation-effect head trained under a realistic protocol. The two need not agree, and on TIM and CASP3 they visibly do not: the layers most linearly aligned with a structural representation are also the deepest, and there a linear notion of redundancy understates what a nonlinear fusion head can still extract. Where they do agree (RBD, and partially CI), the agreement is close enough to be practically useful — the CKA peak sits within two layers of the performance peak — which suggests that a cheap representational screen can shortlist candidate layers before any supervised sweep, but cannot replace it.

Limitations. Four proteins and one model scale (ESM-2 650M) cannot support universal claims; the split we observe (one mid-layer, one penultimate, two last-layer optima) is a sample, not a taxonomy, and the optimum may shift with model size, probe capacity, or metric (we used Spearman ρ throughout). CKA is a linear similarity measure and, as the TIM L33 result shows, underestimates non-linear complementarity. The CI Z_{II} stream carries a known extraction artifact (Methods), so its absolute CKA values should be read with caution. Our noise floor was calibrated on this pipeline's hardware and protocol; other stacks should re-derive it. Finally, our study is silent on zero-shot scoring, where the pre-training objective directly defines the read-out and different considerations apply.

4 Methods

4.1 Datasets

Four single-point DMS datasets were used (Table 2): SARS-CoV-2 RBD (ACE2-binding affinity, `bind_avg`) [9]; human caspase-3 (CASP3) fluorescence activity [10], restricted to the 244-chain

Table 2: Datasets. N = number of mutants scored; positions = unique mutated positions used for position-held-out splitting.

System	Assay	N	Positions	Length	Source
RBD	ACE2 binding	3998	201	201	Starr et al. 2020 [9]
CASP3	Protease activity	1469	244	244 (homodimer 488)	Roychowdhury & Romero 2022 [10]
TIM	Organismal fitness	1519	80	254	Chan et al. 2017 [11]
CI	Expression (FACS)	351	59	237	Li et al. 2019 [12]

positions as in the source pipeline; the *Thermus thermophilus* TrpC TIM barrel (organismal fitness; ProteinGym `TRPC_THEMA_Chan_2017`) [11, 3]; and the λ cI repressor high-expression assay (ProteinGym `RPC1_LAMBD_Li_2019_high-expression`) [12, 3]. Labels were z-scored within each dataset.

4.2 Representations

For every mutant we ran a single-sequence forward pass of ESM-2 650M (`esm2_t33_650M_UR50D`) [2] and stored all 33 transformer-layer hidden states as $[L, 1280]$ tensors (per-mutant, not mutation-delta). The structural stream Z_{II} is the 128-dimensional pair representation of RoseTTAFold3 extracted without any MSA input [8], used as per-residue row slices (RBD/TIM: row grid; CASP3: the mutated row of the 488×488 homodimer pair block, first monomer). The CI Z_{II} extraction uses a legacy indexing convention that is semantically offset but numerically close to the corrected version (project audit: corrected-vs-artifacted baselines differ by ≈ 0.01); we flag all CI CKA values accordingly.

4.3 Probe heads and training

The single-stream (SP) head is a small transformer probe: Linear $1280 \to 256$, sinusoidal positional encoding, a 3-layer Transformer encoder (8 heads, feed-forward 1024, dropout 0.1), an output MLP ($256 \to 512 \to 128$), attention pooling over residues, and a final MLP ($128 \to 64 \to 1$). The fusion (XAttn) head encodes each modality with such an encoder ($\to$ 128 dimensions), applies a per-residue symmetric 2-token cross-attention (1 layer, 4 heads) between the ESM and Z_{II} token at each position, concatenates the streams ($[L, 256]$), and passes them through a 2-layer joint Transformer (8 heads), attention pooling, and an MLP ($256 \to 128 \to 1$). PLM and RF3 weights are frozen; only heads train.

Inputs are standardized per position and per channel (z-score over training samples), which we found mandatory to preserve the ESM channel-variance hierarchy. Splits hold out 30% of mutated *positions* (not mutants) to prevent positional leakage; three split seeds (42–44) and two weight initializations (7, 107) give 6 runs per layer per architecture, shared and paired between SP and XAttn. Training: 250 epochs, AdamW (lr 10^{-4}, weight decay 0.05), cosine schedule, batch size 128 (32 for CI), MSE loss on z-scored labels. The reported metric is best-epoch validation Spearman ρ.

4.4 Noise floor

Identical cells re-run across processes and GPUs in this pipeline differ by up to 0.021 in validation ρ; we therefore adopt ± 0.02 as a reproducibility floor and treat any difference inside it as null. This floor is stated a priori and applied uniformly, including to our own preferred narratives (e.g. the CASP3 L33 fusion marginal $+0.007$ is reported as zero).

4.5 Representational analyses

Linear CKA [7] between each ESM-2 layer and Z_{II} was computed on subsampled (mutant, position) pairs ($\leq$ 300 mutants $\times$ 20 positions per mutant), alongside ridge-regression R^2 (ESM $\rightarrow Z_{II}$, 80/20 split, $\lambda = 100$). Effective rank was measured on the covariance of per-residue vectors as (i) the number of singular values capturing 90% of total variance (eff90) and (ii) the participation ratio $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$.

Data availability

All sweep results are archived as per-protein JSON result files from the four experiment series underlying this study (E74, E75, E75b, E77); every number quoted in the text is mapped to its source file in the accompanying `TRACEABILITY.md`. Analysis scripts and all archived results are publicly available at https://github.com/Lizon1c/esm2-layer-audit-dms.

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